

8. Control Head Up and Down

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8.1 Experiment Objective

This course focuses on learning how to control the robot's head up and head down functions.

8.2 Experiment Preparation

This course involves the following function from the Muto hexapod robot's Python library:

head_move(level): Controls the robot's head up and down movement. The value range for `level` is [0-10], with a default of 5.

8.3 Experiment Procedure

Open the JupyterLab client and navigate to the code path:

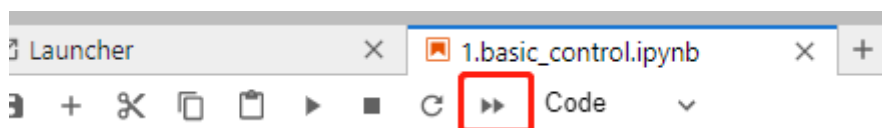
```
muto/Samples/Control/8.control_head.ipynb
```

By default, `g_ENABLE_CHINESE` is `False`. If you need to display Chinese, please set `g_ENABLE_CHINESE=True`.

```
# Chinese switch. The default value is English
g_ENABLE_CHINESE = False

Name_widgets = {
    'End': ("End", "Finish")
}
```

Click 'Run All Cells', then scroll to the bottom to see the generated controls.



Operate the different controls to manage their respective functions.



Each time you slide the Head slider, when the slider's value changes, it will immediately control the robot's head to move up or down.

```
def on_slider_head(value):  
    global g_start  
    if g_start:  
        g_bot.head_move(value)  
        print("head:", value)  
    else:  
        g_start = True
```

8.4 Experiment Summary

In this experiment, JupyterLab controls are used to manage the hexapod robot's head up and down functions.

Note: The head up and down functions cannot be combined with other functions.



To exit the program, press the 'End' button.